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A semantic-enhanced multi-modal remote sensing foundation model for Earth observation

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Remote sensing foundation models, pretrained on massive remote sensing data, have shown impressive performance in several Earth observation (EO) tasks. These models usually use single-modal temporal data for pretraining, which is insufficient for multi-modal applications. Moreover, these models require a considerable number of samples for fine-tuning in downstream tasks, posing challenges in time-sensitive scenarios, such as rapid flood mapping. We present SkySense++, a multi-modal remote sensing foundation model for diverse EO tasks. SkySense++ has a factorized architecture to accommodate multi-modal images acquired by diverse sensors. We adopt progressive pretraining, which involves two stages, on meticulously curated datasets of 27 million multi-modal remote sensing images. The first representation-enhanced pretraining stage uses multi-granularity contrastive learning to obtain general representations. The second semantic-enhanced pretraining stage leverages masked semantic learning to learn semantically enriched representations, enabling few-shot capabilities. This ability allows the model to handle unseen tasks with minimal labelled data, alleviating the need for fine-tuning on extensive annotated data. SkySense++ demonstrates consistent improvements in classification, detection and segmentation over previous state-of-the-art models across 12 EO tasks in 7 domains: agriculture, forestry, oceanography, atmosphere, biology, land surveying and disaster management. This generalizability may lead to a new chapter of remote sensing foundation model applications for EO tasks at scale.

Earth observation (EO), underpinned by remote sensing imagery, has facilitated the monitoring of our planet's static and dynamic processes from a macroscopic perspective[1–3]. Remote sensing images (RSIs) from diverse satellite platforms with different imaging mechanisms, such as optical, near or short-wave infrared and radar signals, provide complementary observations of the Earth characterized by various temporal, spatial and spectral resolutions[4]. The heterogeneity and complexity of this spatio-temporal-spectral distribution pose fundamental challenges to EO[5]. In the past decade, deep learning has demonstrated its potential across a multitude of EO tasks[6]. Nevertheless, training deep models demands large-scale task-specific labelled data, which seldom generalize well across diverse EO applications[7]. Recently, foundation models have exhibited notable generalizability in many domains[8–12]. In particular, remote sensing foundation models (RSFMs)[13–22], pretrained on extensive RSIs, can extract generic features to enable their adaptation to diverse downstream tasks through

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fine-tuning. This pretraining and fine-tuning paradigm has become the common practice for developing RSFMs. Pretraining an RSFM typically involves self-supervised pretraining, during which the model leverages input data to construct learning objectives without manual labels. Specifically, existing RSFMs[13–22] consume vast amounts of unlabelled RSIs by using contrastive learning and masked image modelling[23,24] during pretraining.

Although these RSFMs have demonstrated efficacy on certain EO tasks, their ability to generalize across EO tasks is constrained by three limitations: (1) Limited support for multiple modalities and temporal inputs: Existing single-modal RSFMs, such as SatMAE[13] and Scale-MAE[18], are trained exclusively on single-modal RSIs. Therefore, they are incapable of processing multi-modal inputs in downstream EO tasks, such as tree species classification[25]. Some multi-modal RSFMs, such as DOFA[19] and CROMA[22], support multi-modal images but lack support for multi-temporal inputs. This limitation restricts their application in time-sensitive EO tasks that require multi-temporal inputs, such as change detection[26]. (2) Limited few-shot capability: Current RSFMs primarily extract general representations and then adapt them to downstream tasks by fine-tuning. The efforts to collect and label sufficient data for fine-tuning are often laborious and time-consuming, making these RSFMs unsuitable for urgent EO tasks with limited labelled data, such as rapid flood mapping. (3) Limited use of semantic information: Most RSFMs are pretrained by self-supervised learning. These methods use only unlabelled data for training and do not use semantic information from labelled data. Recently, an attempt was made to pretrain MTP[27] on labelled datasets through multi-task learning, although only optical three-band (RGB) images were explored. Real-world multi-modal datasets vary dramatically in modality and annotation granularity, which pose challenges for data management and use. Furthermore, the use of landscape semantics is complicated by the semantic consistency of applying the same landscape categories across diverse scenes, like 'river' by the sea or in the desert, and the subtle distinctions among similar landscape categories, like swamps and wetlands.

Therefore, we propose three strategies to address these limitations, respectively, which are developed within our recent RSFM, SkySense[28].

Spatio-temporal-modality decoupling

RSFMs should be flexible enough to accommodate input RSIs with different combinations of modalities and spatio-temporal resolutions. Thus, we employ a spatio-temporal-modality decoupling strategy introduced in ref. 28. Features from each modality are first extracted independently and simultaneously. These are then fused temporally to obtain a general representation. In this way, the characteristics of different modalities, for example, optical images with rich textures or synthetic aperture radar (SAR) images robust to weather variability, and their availability are taken into account in flexibly integrating multi-modal and multi-temporal features.

In-context learning

In-context learning was first introduced by GPT-3[29]. A series of natural language processing tasks were formulated as text completion tasks given certain prompts without fine-tuning. Similarly, RSFMs with in-context learning capabilities should be able to make predictions for test images given a few labelled prompt images, thus largely alleviating the need for fine-tuning.

In vision applications, in-context learning has been implemented for RGB images[30–32] as a contextual colouring procedure. The network is trained to predict the colour (the category label) of a masked image region from its surrounding area. This is non-trivial to extend to RSIs, which have several channels and modalities. Therefore, we propose a semantic bank-based strategy for incorporating in-context learning into multi-modal remote sensing tasks, such that continuous colour regression is turned into discrete index segmentation. In this paper, a 'semantic bank' denotes a set of learnable features used to encode the categories of segmentation maps. The same category across different modalities is encoded using features from the same index within its respective semantic bank. Thus, in-context learning is accomplished by segmenting the mask at the assigned indices within the context. For instance, to learn the 'river' category, the pixel-level labels are masked out randomly. The unmasked regions of 'river' are encoded into features of different modalities using features of the multi-modal semantic banks. In-context learning is enabled by segmenting 'river' given the contextual neighbour image regions and the unmasked annotations.

Progressive training

RSFMs should consume both multi-modal unlabelled data and diverse labelled data in pretraining. Thus, we employ a two-stage training strategy: self-supervised learning on extensive unlabelled data to obtain generic representations followed by semantic-enhanced pretraining on large-scale labelled data. The pretraining leverages the semantic consistency of pixels within the same landscape or object categories as well as the semantic inconsistency of pixels among different categories. Instead of learning these patterns separately from individual samples, samples containing pixels in the same category are concatenated in a multi-sample collaborative scheme. This approach allows in-context learning to explore diverse nearby contexts of different samples within a single category, such as 'river' in different geo-locations, thus enhancing the learning of both semantic consistency and inconsistency.

To instantiate these three strategies, we propose SkySense++, a multi-modal RSFM with billion-scale parameters. SkySense++ is an extension of the previous SkySense[28] with new semantic-enhanced in-context learning capabilities. SkySense++, as depicted in Fig. 1a, integrates three remote sensing modalities: high-resolution (HR) optical, multi-spectral (MS) optical and SAR. It supports data from over 11 satellite platforms, including Sentinel-2, Sentinel-1 and Gaofen-2.

SkySense++ inherits the spatio-temporal-modality decoupling architecture of SkySense. Beyond SkySense, which uses only unlabelled multi-modal data in self-supervised pretraining, SkySense++ uses

Fig. 1 | Overview of SkySense++. a, SkySense++ is trained on two pretraining datasets: RS-Rep and RS-Sem. These have RSIs of 3 modalities collected from 11 satellites covering large geo-regions globally. **b**, SkySense++ is pretrained in two stages. During the representation-enhanced pretraining, SkySense++ uses multi-granularity contrastive learning to derive general representations from the RS-Rep dataset. In the semantic-enhanced pretraining, SkySense++ leverages masked semantic learning to integrate the semantics of category labels from the RS-Sem dataset. SkySense++ uses a spatio-temporal-modality decoupling architecture to consume large-scale multi-modal temporal RSIs. **c**, SkySense++ can perform 12 kinds of downstream EO tasks in seven domains: agriculture, forestry, oceanography, atmosphere, biology, land surveying and disaster management. Aug, augmentation; EMA, exponential moving average; ViT, vision transformer; $x_{HR}/x_{MS}/x_{SAR}$, multi-modal temporal RSIs for HR optical/MS optical/SAR modalities; $g_{HR}/g_{MS}/g_{SAR}$, spatial encoders for HR optical/MS optical/SAR modalities; $F_{HR}/F_{MS}/F_{SAR}$, features of HR optical/MS optical/SAR modalities; F_T, initial multi-modal temporal feature; P_{DPTE}, date-specific temporal positional encoding; F_e, extra token; F_T^{cat}, concatenated multi-modal temporal feature; F_{fus}, multi-modal spatio-temporal feature. Spot-5 data in **a** from ref. 81. Satellite image for wildfire monitoring in **c** from Copernicus Browser (https://browser.dataspace.copernicus.eu). Panels adapted with permission from: **a** (Sentinel-1, Sentinel-2), **b** (satellite images), **c** (crop classification), ref. 28, IEEE; **a** (Gaofen-2), ref. 73 under a Creative Commons license CC BY 4.0; **c** (flood monitoring), ref. 41 under a Creative Commons license CC BY 4.0; **c** (landslide mapping), ref. 43, Elsevier; **c** (oil spill segmentation), ref. 36 under a Creative Commons license CC BY 4.0; **c** (forest monitoring), ref. 35 under a Creative Commons license CC BY 4.0. Planet image in **a** reproduced from ref. 79 under a Creative Commons license CC BY-SA 4.0. Credit: icons in **c**, FontAwesome (https://github.com/FortAwesome/Font-Awesome) under a Creative Commons license CC BY 4.0.

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both unlabelled and labelled multi-modal data for pretraining. This is achieved through a progressive pretraining approach in two stages: representation-enhanced pretraining and semantic-enhanced pretraining, as shown in Fig. 1b. During the first stage, SkySense++ uses vast numbers of samples from our RS-Representation (RS-Rep) dataset to learn general multi-modal representation by multi-granularity contrastive learning, following SkySense. In the second stage, SkySense++ uses the new masked semantic learning (MSL) to harness the semantics of

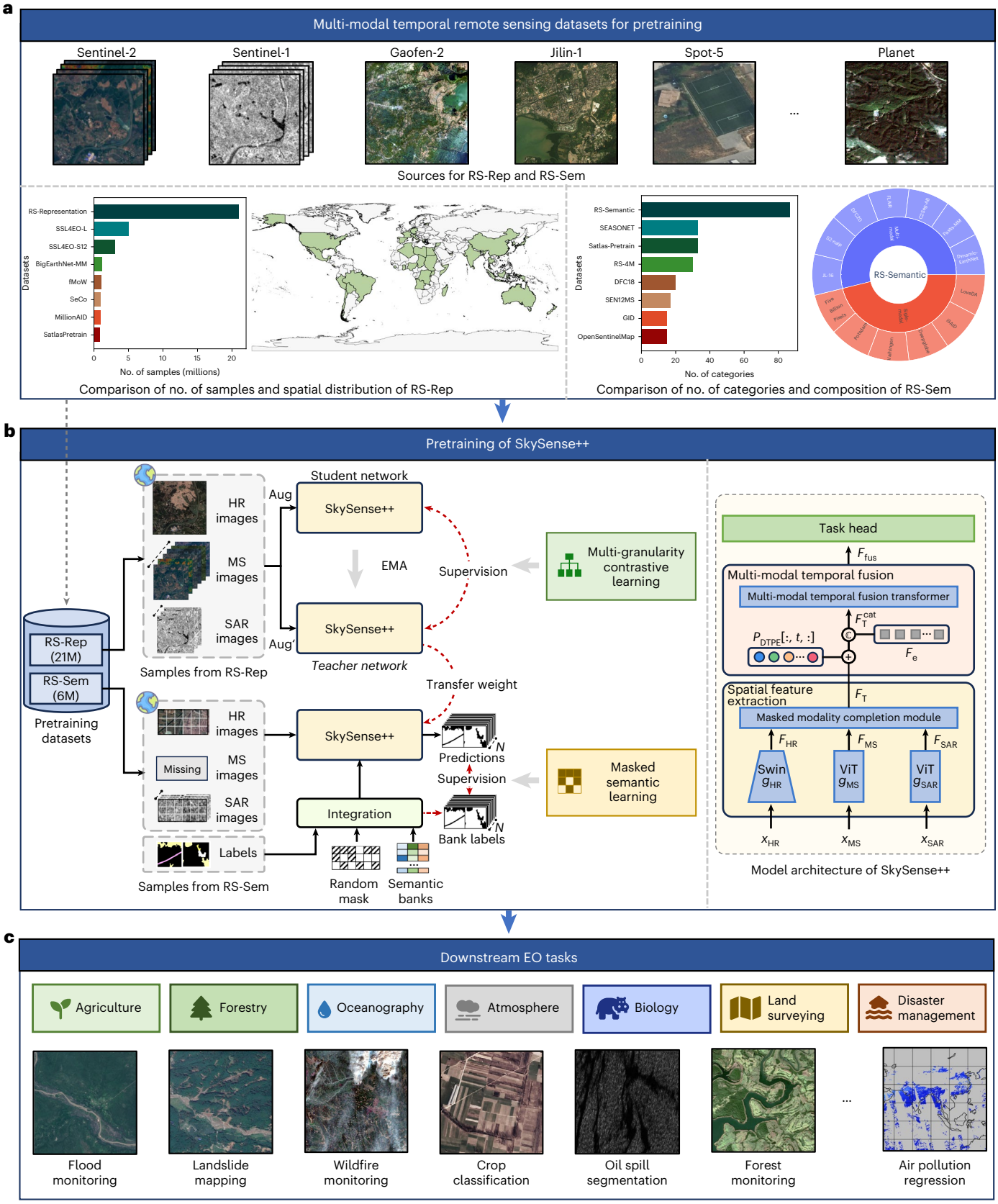

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Table 1 | Fine-tuning performance of different models on 12 types of EO tasks across the 7 domains of agriculture, forestry, oceanography, atmosphere, biology, land surveying and disaster management

Domain	Task type	Dataset	Metrics	Optical RSFM						SAR RSFM	Multi-modal RSFM			
				SatMAE	CMID	CACo	Satlas	GFM	Scale-MAE	SAR-JEPA	CROMA	DOFA	Sky Sense	Sky Sense++
Agriculture	Crop classification	Germany	mIoU (↑)	80.59	75.89	76.06	73.69	76.14	83.71	–	84.18	53.32	85.99	**87.14**
			mF1 (↑)	88.74	85.59	85.70	84.46	85.78	90.86	–	91.51	68.60	92.28	**92.99**
Forestry	Tree species classification	Tree-sat	mAP (↑)	69.27	62.35	66.98	65.35	71.76	70.37	–	69.65	67.21	73.18	**73.45**
			mF1 (↑)	65.42	57.40	62.27	60.04	66.14	67.26	–	65.81	62.63	69.41	**69.91**
	Deforestation segmentation	Atlantic	IoU (↑)	85.48	87.25	86.65	89.14	89.45	84.15	–	84.79	85.63	86.28	**90.57**
			F_1 (↑)	92.17	93.19	92.85	94.26	94.43	91.39	–	92.25	91.77	91.92	**95.05**
Oceanography	Oil spill segmentation	SOS	IoU (↑)	–	–	–	–	–	–	73.12	73.70	73.09	74.87	**75.33**
			OA (↑)	–	–	–	–	–	–	82.41	84.19	84.33	85.09	**87.94**
Atmosphere	Air pollution regression	3pollution	MSE (↓)	0.70	2.11	1.99	0.79	0.72	0.59	–	0.58	0.59	0.60	**0.50**
Biology	Wildlife detection	Kenya	mAP_{50} (↑)	76.00	67.70	55.90	68.70	71.70	76.30	–	57.20	59.30	84.00	**84.50**
			mAP (↑)	38.50	32.30	26.30	32.50	37.60	40.10	–	24.30	26.50	47.50	**48.00**
Land surveying	LULC mapping	C2Seg-BW	mIoU (↑)	34.94	34.78	34.04	34.52	37.47	32.87	–	32.31	34.22	39.24	**39.74**
			mF1 (↑)	49.24	48.82	48.01	48.54	52.06	46.76	–	45.71	48.67	53.76	**54.40**
	Change detection	dsifn-cd	IoU (↑)	48.34	40.52	42.36	36.05	52.60	51.26	–	42.38	39.68	55.20	**59.00**
			F_1 (↑)	65.17	57.67	59.51	53.00	68.94	67.78	–	59.53	56.82	71.13	**74.21**
Disaster management	Flood monitoring	c2smsfloods	mIoU (↑)	76.77	76.90	78.09	79.80	75.61	75.42	–	74.72	73.32	79.93	**80.41**
			mF1 (↑)	86.44	86.52	87.33	88.47	85.68	85.55	–	85.06	84.03	88.51	**88.73**
	Wildfire monitoring	CABUAR	IoU (↑)	69.43	71.55	72.57	73.30	74.90	70.52	–	69.31	69.54	73.45	**76.11**
			F_1 (↑)	81.95	83.42	84.10	84.59	85.65	82.71	–	81.88	82.03	84.70	**86.43**
	Landslide mapping	GVLM	IoU (↑)	40.44	28.56	31.04	35.00	50.42	51.58	–	37.47	46.99	54.91	**56.51**
			F_1 (↑)	57.59	44.43	47.37	51.85	60.11	62.24	–	54.51	63.93	71.08	**72.21**
	Building damage assessment	xBD	mIoU (↑)	60.27	40.20	33.53	53.44	63.01	61.33	–	49.31	43.01	64.96	**65.31**
			mF1 (↑)	72.59	50.77	40.77	65.77	75.29	73.66	–	76.47	55.14	76.89	**77.24**

Each method has undergone comprehensive fine-tuning using identical task-specific training datasets. These methods are categorized into optical, SAR and multi-modal RSFMs according to their input modalities. Notes: A dash indicates that the method is not compatible with the modalities of the dataset. ↑ indicates that higher values signify better performance, whereas ↓ denotes that lower values indicate better performance. Bold values denote the best results for the current task.

category labels from our RS-Semantic (RS-Sem) dataset. The category labels of some random regions are masked out to force the network to recover the semantics of these masked regions. Furthermore, to cope with samples with incomplete modalities, we propose a new masked modality completion module that drops out features of some modalities and learns to reconstruct them, thereby improving training stability and testing robustness for missing modalities. Notably, SkySense++ can handle unseen tasks using only a few annotated samples without fine-tuning, which enhances its adaptability to diverse EO tasks beyond SkySense.

As illustrated in Fig. 1c, SkySense++ is evaluated on 12 typical EO tasks in 7 domains: agriculture, forestry, oceanography, atmosphere, biology, land surveying and disaster management. These tasks encompass a diverse range of applications, from flood monitoring to oil spill segmentation, which provide a practical assessment of RSFMs at scale. Moreover, the few-shot capability of SkySense++ is evaluated on emergent tasks, such as rapid flood mapping. SkySense++ demonstrates a notable improvement over other RSFMs, leading by 4.79% in the mean average precision (mAP) of fine-tuning classification and detection tasks and 14.08% in the mean intersection over union (mIoU) of few-shot segmentation tasks. These enhancements in EO tasks enable precise disaster monitoring and rapid response, enhance environmental conservation efforts, and optimize agricultural and forestry management, among other critical applications. Collectively, SkySense++ demonstrates a step forward for RSFMs to advance sustainable development and enhance societal well-being.

Results

We develop SkySense++, an RSFM applicable to diverse EO tasks. The model is pretrained on two extensive datasets: RS-Rep and RS-Sem. RS-Rep comprises 21 million multi-modal, geographically aligned, unlabelled RSIs from 40 countries and regions, covering an area of 8.78 million km^2. RS-Sem incorporates 398 billion labelled pixels annotated for RSIs with 3 modalities from 11 satellite platforms, annotated into 87 pixel-level categories, including landscape categories such as 'river', 'farmland' or 'building' and object categories such as 'vehicle'

or 'airplane'. Specifically, each annotation in RS-Sem is associated with up to 41 multi-temporal images. These labelled pixels are equivalent to 6 million patches of 256 × 256 pixels. The details of these datasets are illustrated in Fig. 1a and presented in Section 'Pretraining dataset'.

SkySense++ comprises a total of about 2 billion parameters. The two-staged pretraining process for SkySense++ ran on 80 A100-80GB GPUs over 336 h. For a multi-modal sample where the HR optical image had 512 × 512 pixels, the average inference time was 469 ms using an A100-80G GPU. The details are elaborated in Section 'Pretraining of SkySense++'. To evaluate the generalizability, we employ a new EO benchmark, as described in Section 'EO benchmark'. This benchmark evaluates 11 RSFMs in 7 domains with 12 types of EO tasks, which, to the best of our knowledge, provides the most extensive assessment of the practical applicability of RSFMs to date.

EO benchmark

Our EO benchmark is curated to evaluate the performance of RSFMs in practical EO applications. This benchmark encompasses 12 typical tasks in 7 critical domains: agriculture, forestry, oceanography, atmosphere, biology, land surveying and disaster management. Details are presented in Extended Data Table 1. In the agriculture domain, we set up a crop classification task with the widely used Germany dataset[33]. This task is critical for enhancing agricultural productivity and efficiency and for sustainable agricultural development. In the forestry domain, we implement tree species classification and deforestation segmentation tasks that benefit scientific conservation using the recent large-scale TreeSatAI-TS dataset[34] and the Atlantic dataset[35]. In the oceanography domain, we set up a marine oil spill segmentation task, which is vital for environmental protection, emergency management and public health, using the widely adopted SOS dataset[36]. In the atmosphere domain, we adopt an air pollution regression task on the multi-modal 3pollution dataset[37] to aid ecological conservation and promote human health. In the biology domain, we set up a wildlife detection task on the Kenya dataset[38] to support wildlife conservation and promote sustainable ecosystem management. In the land surveying domain, we design land-use and land-cover (LULC) mapping and land change detection tasks using the well-known C2Seg-BW[39] and dsifn-cd[40] datasets. These tasks promote environmental management, urban planning and resource management. In the disaster management domain, we implement tasks for monitoring floods and wildfires, mapping landslides and assessing building damage. These tasks cover the monitoring of disasters and evaluating disaster severity, which are critical for disaster prevention, mitigation and public safety. We use the c2smsfloods[41], CABUAR[42], GLVM[43] and xBD[44] datasets due to their scale and credibility. This benchmark on 12 EO tasks in 7 domains incorporates RSIs from 9 satellite platforms, with a maximum time-series length of 70 images at a single geographic location. It includes ten different modality and temporal sequence combinations, which well represent the complicated availability of multi-modal and temporal sequence data in real EO tasks.

SkySense++ exhibits excellent performance across diverse EO tasks

We compare SkySense++ with ten representative RSFMs: SatMAE[13], CMID[14], CACo[15], Satlas[16], GFM[17], Scale-MAE[18], DOFA[19], DINO-MM[21], SAR-JEPA[20] and CROMA[22]. Each RSFM is fully fine-tuned on the task-specific dataset and evaluated by respective metrics. For segmentation tasks, we use mIoU along with either the mean F_1 score (mF1) or overall accuracy (OA). In detection tasks, mAP and mAP at a 50% intersection over union (IoU) threshold (mAP_{50}) are computed. Both mAP and mF1 are used for multi-label classification tasks. The mean squared error (MSE) is used for regression tasks. Table 1 provides a quantitative comparison of performance across 12 kind of tasks in our benchmark. These results validate the effectiveness of SkySense++. The domain-specific analyses of the experimental results are as follows.

Agriculture. We evaluate SkySense++ on the Germany dataset[33] for crop classification. As all other RSFMs, except Satlas, do not support multi-temporal inputs of this dataset, we use the fusion network in SkySense++ as the fusion module for other RSFMs to ensure a fair comparison. SkySense++ achieves an appreciable pixel-level classification performance in mIoU and mF1, with at least a 2.96% improvement in mIoU and 1.48% in mF1 over all other RSFMs.

Forestry. We evaluate SkySense++ on the TreeSatAI-TS dataset[45] for tree species classification. Unlike the pixel-level classification in the Germany dataset, TreeSatAI-TS sets tree species classification as an image-level multi-label classification task. Each RSFM serves as a feature extractor connected to a linear layer to produce the final results. We evaluate optical RSFMs using HR optical RSIs, as all optical RSFMs support only single-modal optical inputs. SkySense++ surpasses all other RSFMs by at least 1.69% in mIoU and 0.50% in mF1. Additionally, SkySense++ is evaluated on the Atlantic dataset[45] for the deforestation segmentation task and outperforms other RSFMs by 1.12% in IoU and 0.62% in F_1 (Table 1). These findings indicate the potential of SkySense++ for EO tasks in the forestry domain.

Oceanography. We evaluate SkySense++ on the SOS dataset[36] for oil spill segmentation. This pixel-level classification task is used with an overall framework like that used for crop classification. We compare SkySense++ with RSFMs compatible with SAR images. SkySense++ outperforms other RSFMs by at least 1.63% for IoU and 3.75% for OA, as shown in Table 1, highlighting the benefits of using SAR inputs to handle marine EO tasks.

Atmosphere. We evaluate SkySense++ on the 3pollution dataset[37] for air pollution regression. In this task, the MS images from Sentinel-2 are used to complement Sentinel-5P data for the regression of NO_2 concentrations. We combine geospatially aligned Sentinel-5P and Sentinel-2 images along the channel dimension as input and compare the results with those from optical and multi-modal RSFMs. This experiment assesses the ability of RSFMs to generalize to modalities on which they have not been pretrained, as none of these RSFMs are pretrained on the Sentinel-5P images. SkySense++ achieves the best performance on MSE among all other RSFMs, with an error rate reduction of at least 0.09, equivalent to an impressive relative decrease of 18%, testifying the robust modality generalizability of SkySense++.

Biology. We evaluate SkySense++ on the Kenya dataset[38] for wildlife detection, focusing on monitoring elephants, zebras and giraffes. We use the widely adopted Faster-RCNN[46] as the detection framework in the evaluation. SkySense++ improves both mAP and mAP_{50} substantially compared to other RSFMs, with an improvement of at least 8.20% in mAP_{50} and 7.90% in mAP. This validates the effective object-level interpretation capability of SkySense++.

Land surveying. We evaluate SkySense++ on two tasks: LULC mapping on the C2Seg-BW dataset[39] and change detection on the dsifn-cd dataset[40]. Despite not being pretrained on images from Gaofen-3 and Gaofen-5, SkySense++ outperforms competing RSFMs by at least 2.00% in mIoU and 2.34% in mF1 in LULC mapping. These findings verify the generalizability of SkySense++ across images from Gaofen-3 and Gaofen-5, despite their discrepancy from the pretraining data. For change detection, we use a shared feature extractor to derive and compare features from two temporal phases, feeding the differential features into a segmentation head to output the final change maps. SkySense++ exhibits a notable edge over other RSFMs, with at least a 3.80% higher IoU and a 3.08% increase in F_1.

Disaster management. We evaluate SkySense++ on four tasks: flood monitoring, wildfire monitoring, landslide mapping and build

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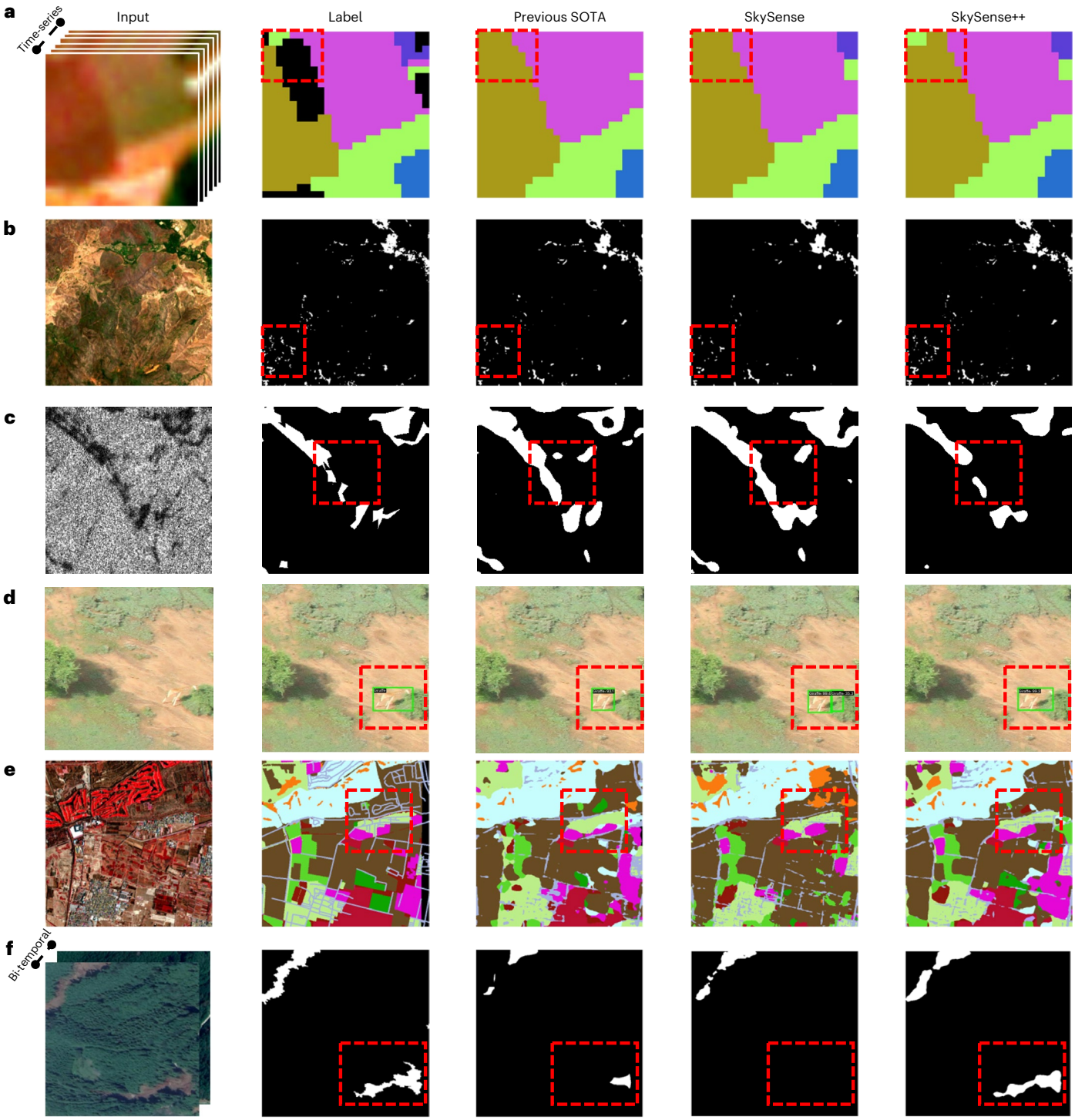

Fig. 2 | Visual comparison of the previous state-of-the-art method, SkySense and SkySense++ in diverse domains. a, Segmentation maps for the Germany dataset in the agriculture domain. **b**, Segmentation maps for the Atlantic dataset in the forestry domain. **c**, Segmentation maps for the SOS dataset im the oceanography domain. **d**, Bounding boxes for the Kenya dataset in the biology domain. **e**, Segmentation maps for the C2Seg-BW dataset in the land surveying domain. **f**, Segmentation maps for the GVLM dataset in the disaster management domain. Regions with clear advantages are highlighted with red dashed boxes. SOTA, state of the art. Input images reproduced with permission from: **a**, reproduced with permission from ref. 33, under a Creative Commons license CC BY 4.0 license; **b**, ref. 35, under a Creative Commons license CC BY 4.0 license; **c**, ref. 36, under a Creative Commons license CC BY 4.0 license; **d**, ref. 38, under a Creative Commons license CC0 1.0 license. Input images adapted with permission from: **e**, ref. 39, Elsevier; **f**, ref. 43, Elsevier.

damage assessment. The flood and wildfire monitoring tasks involve generating segmentation maps from a single-temporal RSI. Landslide mapping and building damage assessment require change maps for landslide areas and damaged buildings and, thus, use bi-temporal HR RSIs. SkySense++ improves mIoU by at least 0.61% and mF1 by 0.26% in the flood monitoring task, enhances IoU by 1.21% and F_1 by 1.84% in wildfire monitoring, notably boosts IoU by 4.93% and F_1 by 9.97% in landslide mapping, and increases mIoU by 2.30% and mF1 by 1.95% in building damage assessment. Specifically, compared to the flood and wildfire monitoring tasks using single-temporal data,

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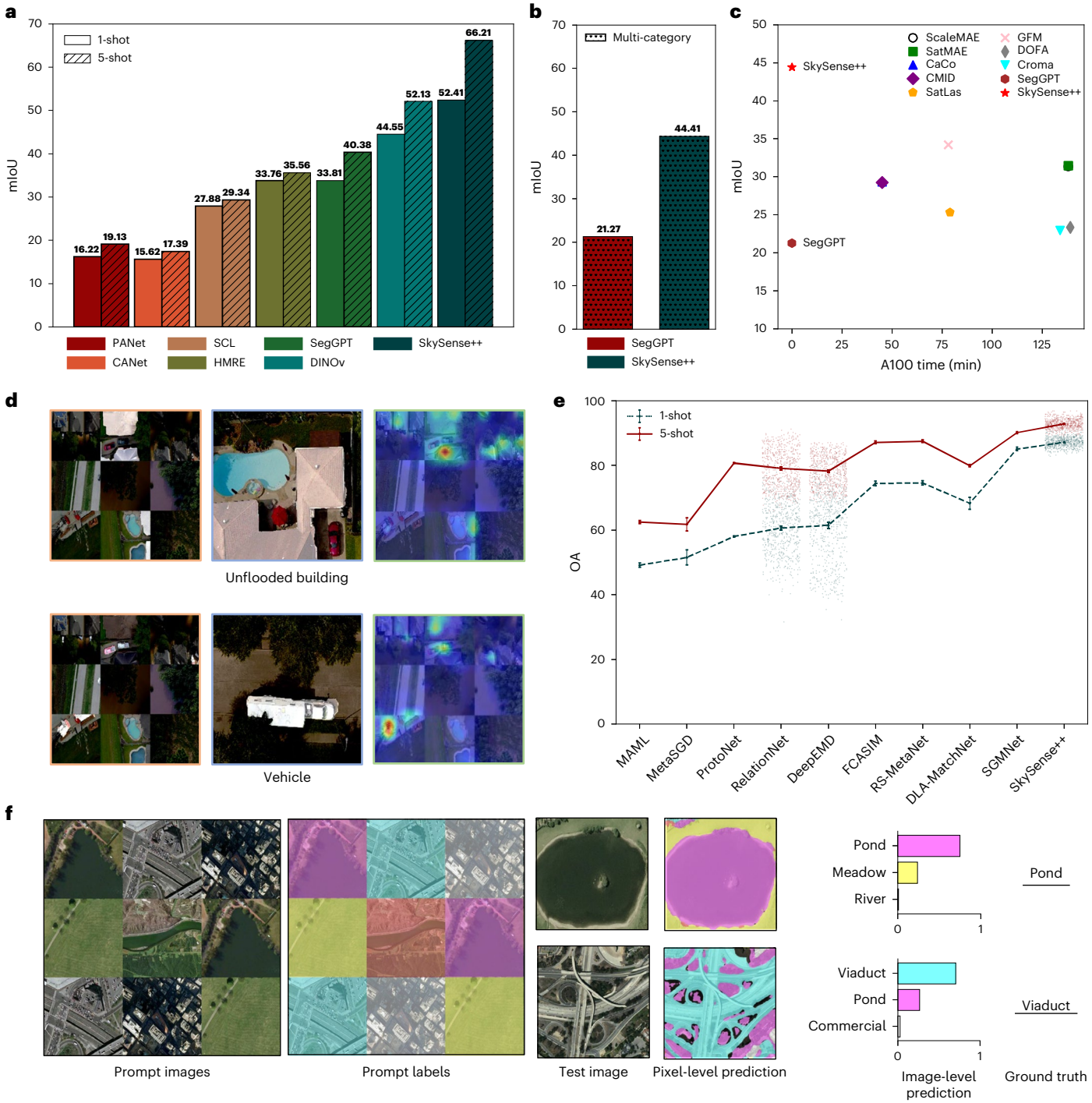

Fig. 3 | Comparison of SkySense++ and other methods in the few-shot tasks. **a**, Comparison of one-shot and five-shot performance as measured by mIoU across various methods on the Flood-3i dataset. SkySense++ outperforms all generalist and specialist models. **b**, Comparison of few-shot segmentation performance as measured by mIoU between SegGPT and SkySense++ using multi-category prompts on the Flood-3i dataset. **c**, Comparison of response times for different foundation models on the Flood-3i dataset. Points closer to the upper left corner indicate faster and more accurate responses. **d**, Correlation analysis between prompt and test features. The orange boxes are the prompt images and the corresponding masks. The blue boxes show the target images and the mask outputs. The green boxes present Grad-CAM[53] visualizations that illustrate the response of the prompt feature in different spatial areas to the predicted class region. The mask is visualized through the bright region attached to the image. **e**, Comparison of one-shot and five-shot performance as measured by OA for various methods with the WHU-RS19 dataset. Following the experimental setting in ref. 63, each method is evaluated with 600 repeated experiments ($n = 600$). The results for MAML, MetaSGD, ProtoNet, FCASIM, RS-MetaNet, DLA-MatchNet and SGMNet are cited in ref. 63. The plot displays the mean values for these experiments, with error bars indicating the 95% confidence interval. **f**, Visualization results for SkySense++ on the WHU-RS19 dataset. Different colours in the prompt labels indicate different semantic embeddings. SkySense++ can provide both image-level and pixel-level predictions. Data in **d** from refs. 50,52. **f** reproduced with permission from ref. 54 under a Creative Commons license CC 0 license.

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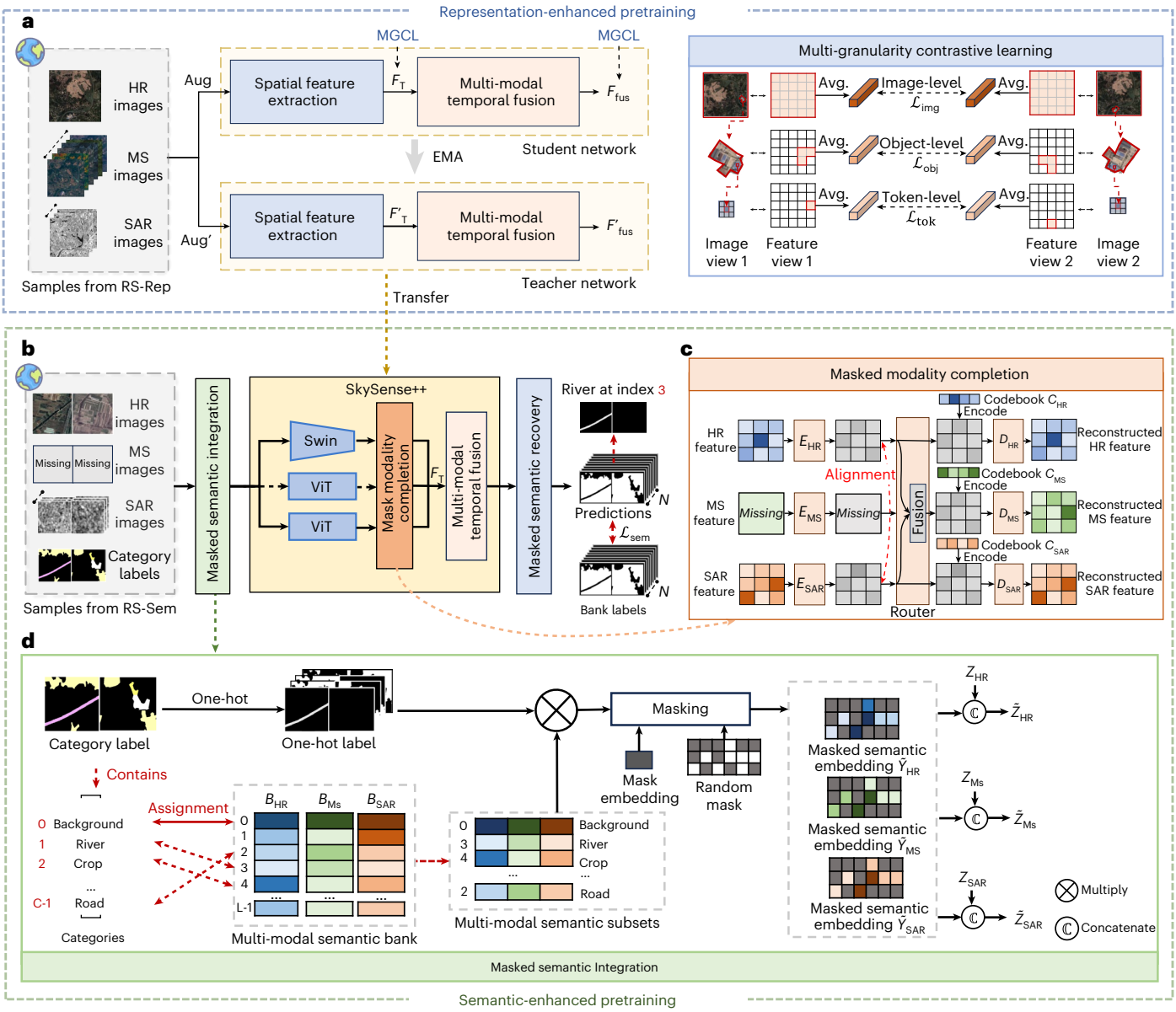

Fig. 4 | Representation-enhanced and semantic-enhanced pretraining. **a**, Representation-enhanced pretraining uses a teacher–student architecture. Within the student branch, multi-granularity contrastive learning facilitates representation learning at several granularities for general EO tasks. **b**, Pipeline of semantic-enhanced pretraining for MSL. **c**, The masked modality completion module drops out the features of some modalities and learns to reconstruct them during pretraining to handle incomplete modalities. **d**, Masked semantic integration uses a multi-modal semantic bank to encode randomly masked labels into masked semantic embeddings. This process provides randomly masked category labels to the network and leads to a new learning objective, that is, to reconstruct the bank labels, facilitating subsequent in-context learning. Avg., average; MGCL, multi-granularity contrastive learning; Swin, Swin Transformer; $\mathcal{L}_{\text{img}}/\mathcal{L}_{\text{obj}}/\mathcal{L}_{\text{tok}}$, image/object/token-level contrastive loss; $\mathcal{L}_{\text{sem}}$, semantic recovery loss. Modality-specific components (HR optical, MS optical, and SAR, with subscripts HR/MS/SAR): E, completion encoders; C, codebooks; D, completion decoders; B, semantic banks; Z, image embeddings; $\tilde{Y}$, masked semantic embeddings; $\tilde{Z}$, masked semantic image embeddings. **a** and **b** adapted with permission from ref. 28, IEEE.

SkySense++ shows clear advantages using bi-temporal data, validating the effectiveness of multi-modal multi-temporal pretraining in disaster management.

Figure 2 presents a qualitative visual comparison of RSFMs across various EO tasks. We select representative visualization results of six EO tasks from different domains. The results indicate that SkySense++ can better segment, detect and recognize landscape or object categories in different RSIs among all the RSFMs. These visual comparisons in Fig. 2a–f indicate that SkySense++ identifies the categories and delineates more accurate category boundaries compared to other methods. Notably, our previous work, SkySense, also achieves sound performance. SkySense++ considerably improves upon SkySense due to the semantic-enhanced in-context learning.

SkySense++ facilitates EO tasks with minimal prompting

We evaluate the few-shot capabilities of SkySense++ on the flood mapping tasks. The comparison methods include both few-shot segmentation methods and generalist models. These few-shot segmentation methods[47–50] often require training on a task-specific training set, which limits their applicability. Differently, generalist models such as SegGPT[32], DINOv[51] and SkySense++ enable direct inference without task-specific fine-tuning.

A quantitative comparison with the Flood-3i dataset[52,50] is presented in Fig. 3a. Notably, SkySense++ with the architecture in Fig. 4 surpasses specialized methods[47–50] by at least 30.65% for mIoU and outperforms the generalist models[32,51] by at least 14.08% for mIoU in the five-shot setting. In Fig. 3a, we adopt the evaluation methodology from ref. 50, where the predictions were obtained by providing the single-category prompts and limiting each inference to predicting only one category. As shown in Fig. 3b, the evaluation results of multi-category inference indicate that SkySense++ achieves an improvement of over 23.14% in mIoU compared to SegGPT. These advantages validate the effectiveness of SkySense++ in flood mapping tasks. The ability to generate accurate segmentation maps with minimal prompts without fine-tuning is critical to time-sensitive EO tasks and greatly eases the deployment for EO applications.

To quantitatively assess this capability, we compare mIoU with respect to response time. We ensure a fair evaluation by conducting all experiments using an identical Nvidia A100 GPU. Using the same prompts as in Fig. 3b, we fine-tune the RSFMs that lack few-shot capability by treating these prompts as a training set for 10,000 iterations. The comparisons in Fig. 3c, response time versus mIoU, reveal that generalist models with few-shot capability consistently deliver in a shorter response time than other RSFMs. SkySense++ achieves the highest mIoU in a low response time, which is critical to rapid development in urgent EO applications.

SkySense++ segments the different categories based on corresponding category prompts. For instance, when segmenting the 'vehicle' in the test image, SkySense++ primarily relies on the 'vehicle' region indicated in the prompt. To validate this, we conduct a correlation analysis between the prompt and test features. Specifically, we use Grad-CAM[53] to display the gradient magnitudes of prompt features used for predicting different categories, highlighting their category-specific contributions to segmentation. The Grad-CAM visualizations in Fig. 3d show that the prompt features of those categories contribute the most to predicting the corresponding categories in the test images, demonstrating the effective in-context learning capability of SkySense++.

We compare the few-shot classification ability of SkySense++ with other methods with the WHU-RS19 dataset[54]. Notably, SkySense++ was not pretrained on image-level labels, so the evaluation was to check its generalizability in a few-shot setting. As illustrated in Fig. 3f, unlike pixel-level prompts used in few-shot segmentation, SkySense++ conducts classification with image-level prompts, where prompt images of different categories are assigned to distinct prompt labels.

For example, the images of a 'pond' and a 'viaduct' are assigned different prompt labels. Using these prompt images and labels, SkySense++ first generates pixel-level predictions for test images and then derives image-level predictions by computing the spatial proportion of each category. The results in Fig. 3e show that SkySense++ achieves the best performance in terms of mean OA among other few-shot classification methods[55–63], leading by over 2% in mean OA for both one-shot and five-shot settings without being trained on task-specific datasets or image-level labels. Although image-level prompt labels are spatially coarse, SkySense++ produces pixel-level predictions that identify the categories and provide high-quality segmentation maps, such as the predictions for 'pond' and 'viaduct' in Fig. 3f.

Discussion

What are the special requirements for RSFMs as foundation models?

A foundation model should serve as a generic feature extractor that can readily be tailored to handle a variety of tasks. This has been accomplished for homogeneous natural images by self-supervised learning on massive unlabelled data[13–22]. By contrast, both the input data and the supported tasks are heterogeneous for RSFMs. The RSIs of different modalities capture distinct spectral ranges of landscapes, which are heterogeneous in nature. Moreover, their availabilities could be dramatically different from diverse satellite platforms. Thus, an RSFM should be flexible and robust in accepting input images with different modalities, spatial and temporal resolutions. Therefore, SkySense++ decouples the feature extraction of different modalities and fuses them temporally by resorting to the spatial alignment among multi-modal images for self-supervised learning. Besides the inputs, the EO tasks are heterogeneous in that they may require pixel-level or image-level recognition or segmentation on different combinations of multi-modal data. Generic features still need to be fine-tuned with substantial labelled task-related data. This motivated us to extend the few-shot capability of RSFMs by pretraining SkySense++ on extensive labelled RSIs. The decoupled modular design and the use of both labelled and unlabelled data make SkySense++ stand out as a generalist for abundant downstream EO tasks.

Should we scale up the RSFM data-wise or model-wise?

The generalizability of foundation models is rooted in the scale of the training data and model size, as illustrated by scaling laws[64]. In SkySense++, the network was scaled up to about 2 billion parameters, pretrained on 21.5 million unlabelled and 6 million labelled multi-modal data, which are all the labelled RSIs we had collected to date. This data-driven scalability empowers the SkySense++ model to generalize well on complicated application scenarios, as validated in Section Results. In fact, both SkySense++ and previous SkySense models have a nearly identical number of parameters. Nevertheless, SkySense++ is trained with an extra 6 million labelled data by semantic-enhanced pretraining. The improvement of SkySense++ over SkySense indicates that model capacity may not be the bottleneck to performance given more training data and better pretraining objectives. Furthermore, we tried to increase the model size to about 3.6 billion parameters, leading to a marginal 0.4% increase in OA in LULC mapping. Our hunch is that for image recognition and segmentation tasks, the current model with 2 billion parameters may be sufficient for now. We will further scale up the number of data and try larger models until we see signs of saturated performance.

Should large language models be integrated into RSFMs?

Large language models (LLMs) have recently achieved tremendous success in many fields[8,12]. Integrating an LLM with an RSFM would easily pump the model size up to even a hundred billion parameters. However, most EO tasks are highly domain-specific visual observation tasks and do not involve heavy textual interactions with people. Moreover, fine-tuning or inference of LLMs generally requires much more computation resources than vision models. Widely used LLMs, such as the Llama series of models[65–67], include models with at least 7 billion parameters, far more than the 2 billion parameters in SkySense++. We have explored integrating a Llama2-7B model into SkySense++, which showed marginal improvements in some image-level recognition tasks, but there was a performance drop on a large portion of pixel-level tasks. Therefore, we do not incorporate an LLM into SkySense++ at present. We believe further investigation is required to achieve a trade-off when integrating LLM to boost overall performance for remote sensing tasks.

Methods

Pretraining dataset

To pretrain SkySense++, we constructed a large-scale multi-modal remote sensing dataset, comprising RS-Rep and RS-Sem, which are depicted in Fig. 1.

RS-Rep is an unlabelled, multi-modal temporal and spatially aligned RSI dataset. It includes images from three modalities: HR optical imagery from WorldView-3, 4 and so on, medium-resolution temporal MS imagery from Sentinel-2, and temporal SAR imagery from Sentinel-1. Although both the HR optical imagery and MS imagery technically belong to the optical modality, we consider them as two

modalities for simplicity due to their differences in spectral characteristics and ground sample distance. The RS-Rep contains 21.5 million training samples. Each sample comprises a static HR optical image with rich texture details, temporal MS imagery with time-series MS data, temporal SAR imagery providing backscatter polarizations, and metadata such as acquisition dates for temporal modelling. As illustrated in the second row of Fig. 1a, RS-Rep contains more RSIs than other self-supervised pretraining datasets[16,68–72]. RS-Rep contains over three times as many images as the largest existing dataset, SSL4EO-L[68].

RS-Sem is a labelled RSI dataset with 398 billion annotated pixels collected from 12 existing datasets[28,39,73–83] plus the newly labelled JL-16 dataset. These annotated pixels are equivalent to 6 million patches of 256 × 256 pixels. Details of the datasets are listed in Extended Data Table 2. RS-Sem incorporates RSIs in 3 modalities from 11 diverse satellite platforms, annotated into 87 landscape or object categories. The comparison in the second row of Fig. 1a indicates that the number of pixel-level annotated categories in RS-Sem is larger than that in other datasets[16,82,84–88]. We establish a standard way to organize these data, which enables consistent access of samples with different modalities in semantic-enhanced pretraining. Specifically, we predefine specific SkySense++ encoders for each image within the multi-modal data samples according to their respective modalities and record this in the metadata. As some modalities of the multi-modal data samples are absent, such as the lack of HR images in C2Seg-AB data samples[39] or the absence of Sentinel-1 images in FLAIR data samples[81], we document the available modalities for each multi-modal sample in the metadata, which is essential for accessing these samples smoothly in semantic-enhanced pretraining.

The input samples for SkySense++ are denoted by x_{HR}, x_{MS} and x_{SAR}, where x_{HR} denotes static HR optical imagery, x_{MS} represents the temporal MS imagery and x_{SAR} refers to the standard-calibrated temporal SAR imagery.

Model architecture

Figure 1b illustrates our proposed network, which features a factorized multi-modal spatio-temporal encoder. It has two parts: spatial feature extraction and multi-modal temporal fusion. In spatial feature extraction, the network independently extracts spatial features from multi-modal temporal RSIs. In multi-modal temporal fusion, these extracted features are fused to generate a multi-modal spatio-temporal representation. This design decouples spatial feature extraction from feature fusion, thereby enhancing the extraction of heterogeneous features from different modalities while facilitating the integration of features across modalities and temporal dimensions.

Spatial feature extraction. The encoder takes multi-modal temporal samples $\{x_{\mathrm{HR}}, x_{\mathrm{MS}}, x_{\mathrm{SAR}}\}$, which are geographically aligned, as inputs. We use the spatial encoders g_{HR}, g_{MS} and g_{SAR} for RSIs from different modalities. The features obtained, $F_i \in \mathbb{R}^{h \times w \times T_i \times d}$, have the same size in the spatial dimension:

$$F_i = g_i(x_i), \qquad i \in \{\mathrm{HR}, \mathrm{MS}, \mathrm{SAR}\}. \tag{1}$$

Here, h and w are the height and width of a feature, T_{HR}, T_{MS} and T_{SAR} represent the temporal lengths of HR optical, temporal MS and temporal SAR images, respectively, and d represents the feature dimension. These features are then passed through the masked modality completion module, denoted as h_{MMC}:

$$F^{\mathrm{M}}_{\mathrm{HR}}, F^{\mathrm{M}}_{\mathrm{MS}}, F^{\mathrm{M}}_{\mathrm{SAR}} = h_{\mathrm{MMC}}(F_{\mathrm{HR}}, F_{\mathrm{MS}}, F_{\mathrm{SAR}}). \tag{2}$$

h_{MMC} reconstructs the features of missing modalities based on the available modalities. This crucial component enhances the robustness of the model to incomplete modal data. Details are provided in Section 'Semantic-enhanced pretraining'. Here $F^{\mathrm{M}}_{\mathrm{HR}}$, $F^{\mathrm{M}}_{\mathrm{MS}}$ and $F^{\mathrm{M}}_{\mathrm{SAR}}$ are the modality-completed features of HR optical, MS and SAR images, respectively. The initial multi-modal temporal feature representations $F_{\mathrm{T}} \in \mathbb{R}^{N_{\mathrm{S}} \times N_{\mathrm{T}} \times d}$ are generated by concatenating all F^{M}_i along the time dimension, where $N_{\mathrm{S}} = hw$ represents the feature size in the spatial dimension and N_{T} represents the total temporal length across all modalities:

$$F_{\mathrm{T}} = \mathrm{Concat}\left[F^{\mathrm{M}}_{\mathrm{HR}}, F^{\mathrm{M}}_{\mathrm{MS}}, F^{\mathrm{M}}_{\mathrm{SAR}}\right]. \tag{3}$$

Multi-modal temporal fusion. To leverage the precise acquisition dates of RSIs, we introduce a date-specific temporal positional encoding $P_{\mathrm{DTPE}} \in \mathbb{R}^{1 \times 365 \times d}$, a learnable parameter that encodes different dates within a year. This encoding is crucial for tasks influenced by seasonal variations. $P_{\mathrm{DTPE}}[:, t, :] \in \mathbb{R}^{1 \times N_{\mathrm{T}} \times d}$ represents the temporal encoding of all RSI acquisition dates, where $t \in \mathbb{R}^{N_{\mathrm{T}}}$ denotes the acquisition dates. This encoding is then added to F_{T} to produce $F^{\mathrm{date}}_{\mathrm{T}}$, a feature representation that explicitly incorporates temporal information. Subsequently, $F^{\mathrm{date}}_{\mathrm{T}}$ is concatenated with another token $F_{\mathrm{e}} \in \mathbb{R}^{N_{\mathrm{S}} \times 1 \times d}$, inspired by the class token in vision transformers[89], which aggregates features across the entire image:

$$\begin{aligned} F^{\mathrm{date}}_{\mathrm{T}} &= F_{\mathrm{T}} + P_{\mathrm{DTPE}}[:, t, :], \\ F^{\mathrm{cat}}_{\mathrm{T}} &= \mathrm{Concat}\left[F_{\mathrm{e}}, F^{\mathrm{date}}_{\mathrm{T}}\right] \in \mathbb{R}^{N_{\mathrm{S}} \times (1+N_{\mathrm{T}}) \times d}. \end{aligned} \tag{4}$$

The concatenated multi-modal temporal feature $F^{\mathrm{cat}}_{\mathrm{T}}$ serves as the input to the multi-modal temporal fusion transformer. This fusion module uses a 24-layer transformer network[90] to integrate multi-modal temporal features, thereby generating a multi-modal spatio-temporal feature $F_{\mathrm{fus}} \in \mathbb{R}^{N_{\mathrm{S}} \times 1 \times d}$.

Pretraining SkySense++

Representation-enhanced pretraining. Figure 4a is an overview of our representation-enhanced pretraining. This pretraining leverages a common teacher–student structure[91] for self-supervised learning. This pretraining uses multi-modal samples derived from RS-Rep. The parameters in the teacher network are dynamically updated using an exponential moving average[92] of the parameters in the student network, thus ensuring a stable learning process.

Multi-granularity contrastive learning. We adopt multi-granularity contrastive learning, a self-supervised learning approach that operates across diverse modalities and spatial granularities. Our method takes $\{x_{\mathrm{HR}}, x_{\mathrm{MS}}, x_{\mathrm{SAR}}\}$ as inputs and applies two sets of random augmentations to generate corresponding view pairs. These augmented views are then processed through the two branches, like the process described in Section 'Model architecture'. As illustrated in Fig. 4a, we establish token-, object- and image-level contrastive learning to progressively learn coarse-to-fine spatial features for various tasks. The fine-grained contrastive learning loss $\mathcal{L}_{\mathrm{FGCL}}$ based on the above principle is defined as:

$$\mathcal{L}_{\mathrm{FGCL}}\left(F_i, F'_i\right) = \sum_{n \in \{\mathrm{tok,obj,img}\}} \mathcal{L}_n\left(F_i, F'_i\right), \tag{5}$$

where F_i represents the feature extracted from the student branch and F'_i denotes the feature extracted from the teacher branch. We use the contrastive loss $\mathcal{L}_{\mathrm{CL}}$ based on self-distillation[91] to learn feature representations by aligning the outputs of a teacher network with those of a student network with the different views of one image, thereby eliminating the need to explicitly define negative samples from other images of the mini-batch. Both F_i and F'_i can be viewed as token-level features. The token-level contrastive learning loss $\mathcal{L}_{\mathrm{tok}}$ is computed by averaging contrastive losses $\mathcal{L}_{\mathrm{CL}}$ across features at the token level. The object-level contrastive learning loss $\mathcal{L}_{\mathrm{obj}}$ is computed by averaging contrastive losses across different feature clusters. More details of the feature cluster process are given in Supplementary Section 'Object-level

contrastive loss'. The image-level contrastive learning loss $\mathcal{L}_{\text{img}}$ is derived by averaging contrastive losses across image-level features, which are derived by average pooling on F_i and F'_i. In the self-distillation framework, the computation of $\mathcal{L}_{\text{tok}}$ relies solely on the features extracted from a single image across different views, with the remaining images in the mini-batch being excluded from this process.

We then formulate the multi-granularity contrastive learning loss $\mathcal{L}_{\text{MGCL}}$ in equation (6). The multi-granularity is reflected in two aspects: spatial and modality. Spatially, it takes into account pixel-, object- and image-level representations. Modality-wise, it applies to both single-modality features F_i and fused multi-modal features F_{fus}:

$$\mathcal{L}_{\text{MGCL}} = \sum_{i\in\{\text{HR,MS,SAR}\}} \mathcal{L}_{\text{FGCL}}\left(F_i, F'_i\right) + \mathcal{L}_{\text{FGCL}}\left(F_{\text{fus}}, F'_{\text{fus}}\right). \tag{6}$$

Semantic-enhanced pretraining. After the representation-enhanced pretraining, the semantic-enhanced pretraining improves the representations to obtain the few-shot capabilities through MSL, which is a major technical contribution beyond our previous SkySense[28]. Unlike masked image modelling[24], which masks and reconstructs random regions of an input image, MSL applies this process to landscape category labels. This approach takes advantage of the extensive category labels, enabling SkySense++ to learn both the consistencies and inconsistencies between categories and to acquire in-context learning capabilities. To achieve this, we introduce a new semantic bank-based method. Using the network architecture in Section 'Model architecture', the SkySense++ model is trained by MSL with three primary schemes: (1) masked semantic integration, which encodes randomly masked category labels as network inputs, (2) masked modality completion, which drops out the features of some modalities and learns to reconstruct them, thereby improving training stability and testing robustness for missing modality, and (3) masked semantic recovery, which reconstructs the category labels of masked pixels or regions from the multi-modal features of their nearby contexts.

Masked semantic integration. To encode pixel-level category labels into the network, we propose masked semantic integration. We will first introduce the network inputs and the multi-modal semantic bank, followed by describing the masked semantic integration process.

Rather than using separate individual images, the inputs in the semantic-enhanced pretraining are organized as in-context image samples $\{X_{\text{HR}}, X_{\text{MS}}, X_{\text{SAR}}\}$ along with their corresponding category label Y from the RS-Sem dataset, as shown in Fig. 4b. One in-context image sample is a set of images that share at least one common category in their category labels. X_{HR}, X_{MS} and X_{SAR} correspond to in-context image samples for the HR optical, MS and SAR modalities, respectively. For example, $X_{\text{HR}} = \{x^{(1)}_{\text{HR}}, x^{(2)}_{\text{HR}}\}$ indicates that the images $x^{(1)}_{\text{HR}}$ and $x^{(2)}_{\text{HR}}$ are in this in-context image sample and that they share at least one category. The category labels Y are segmentation maps that include C categories, where C varies from 5 to 24 per iteration according to the input samples. The in-context image samples from different modalities are independently concatenated and fed into the network to enhance prediction consistency within the same category and inconsistency across different categories, a process we term a multi-sample collaborative scheme.

We employ multi-modal semantic banks to encode the category labels of these in-context image samples of different modalities. In this paper, the term 'semantic bank' denotes a set of learnable features used to encode the category labels. As shown in Fig. 4d, the multi-modal semantic banks comprise three distinct banks for different modalities: $B_{\text{HR}} \in \mathbb{R}^{d_{\text{HR}}\times L}$ for the semantic bank of the HR modality, $B_{\text{MS}} \in \mathbb{R}^{d_{\text{MS}}\times L}$ for the MS modality and $B_{\text{SAR}} \in \mathbb{R}^{d_{\text{SAR}}\times L}$ for the SAR modality, where d_{HR}, d_{MS} and d_{SAR} are feature dimensions of the semantic banks of the corresponding modality. These semantic banks have the same bank length L, which is set to 65 more than the total number of categories in the in-context image samples. Different semantic banks are used to capture the heterogeneity of different modalities and accommodate varying numbers of bands in RSIs from different modalities.

The process for masked semantic integration is shown in Fig. 4d. Category label Y contains a variety of categories, such as 'background', 'river', 'crop' and 'road', indexed by $\{0, 1, \ldots, C-1\}$. C denotes the total number of categories. To handle an unseen landscape category, rather than simply memorizing a fixed sequence of categories, in each training iteration, the categories indexed by $\{1, \ldots, C-1\}$ in Y are randomly assigned to a subset of the multi-modal semantic bank with corresponding indices $\{l_1, \ldots, l_{C-1}\}$. The corresponding indices in the semantic banks for 'background' are fixed to 0. The multi-modal semantic subsets $\{\tilde{B}_{\text{HR}} \in \mathbb{R}^{d_{\text{HR}}\times C}, \tilde{B}_{\text{MS}} \in \mathbb{R}^{d_{\text{MS}}\times C}, \tilde{B}_{\text{SAR}} \in \mathbb{R}^{d_{\text{SAR}}\times C}\}$ are obtained by filtering the features of the indices $\{0, l_1, \ldots, l_{C-1}\}$ from the corresponding semantic banks. Furthermore, given this assignment, the category labels Y are transformed into bank labels Y_{B}, which consist of L channels that match the length of the semantic bank. The channels indexed by $\{0, l_1, \ldots, l_{C-1}\}$ of Y_{B} contain segmentation maps for classes $\{0, 1, \ldots, C-1\}$ of the category labels Y. Although different semantic banks are used for each modality, they are interconnected by using the same indices during the construction of the multi-modal semantic subsets and bank labels.

To encode the category labels, as shown in Fig. 4d, a category label is first converted into a one-hot label with C channels, where each channel represents a different category. The one-hot label is then encoded into semantic embedding $\{Y_{\text{HR}}, Y_{\text{MS}}, Y_{\text{SAR}}\}$ by replacing each pixel of the one-hot label with the corresponding features of the multi-modal semantic subsets. The 'semantic embedding' refers to the features of the label encoded by the semantic bank. This process is implemented through matrix multiplication:

$$Y_i = \text{OneHot}(Y) \times \tilde{B}_i^{\text{T}}, \qquad i \in \{\text{HR, MS, SAR}\}, \tag{7}$$

where OneHot(·) denotes the one-hot encoding. Afterwards, a random mask M_{R} is applied to each Y_i to obtain masked semantic embeddings $\{\tilde{Y}_{\text{HR}}, \tilde{Y}_{\text{MS}}, \tilde{Y}_{\text{SAR}}\}$:

$$\tilde{Y}_i = Y_i(1 - M_{\text{R}}) + Y_i^{\text{M}} M_{\text{R}}, \qquad i \in \{\text{HR, MS, SAR}\}, \tag{8}$$

where Y_i^{M} are the masked embeddings for modality i. Finally, the image embedding Z_i and masked semantic embedding $\tilde{Y}_i$ are concatenated to obtain the masked semantic image embedding $\tilde{Z}_i$:

$$\tilde{Z}_i = \text{Concat}[Z_i; \tilde{Y}_i], \qquad i \in \{\text{HR, MS, SAR}\}, \tag{9}$$

where Concat[·] means the concatenation operation. The image embedding $Z_i = f_{\text{PE}}(X_i)$ is derived by applying the patch embedding function f_{PE} to the input X_i.

Masked modality completion. Not all modalities are always available for either training or testing image samples. The training of RSFM should be flexible enough to take any available modalities as input, whereas inference by RSFM should be robust enough to tolerate missing modalities. Thus, we propose a masked modality completion scheme to reconstruct the features of missing modalities based on the features of the available modalities, as illustrated in Fig. 4c.

The inputs to the module are the features of each modality, namely F_{HR}, F_{MS} and F_{SAR}. The corresponding encoders, E_{HR}, E_{MS} and E_{SAR}, encode these features into a unified latent space and obtain latent features F^{L}_{HR}, F^{L}_{MS} and $F^{\text{L}}_{\text{SAR}}$. The latent features of each modality represent the distribution over the corresponding codebook. It is expected that the latent features from the available modalities will be aligned to support cross-modal feature completion. Therefore, we used the MSE loss $\mathcal{L}_{\text{MSE}}$ to align the features. The latent alignment loss $\mathcal{L}_{\text{latent}}$ is

$$\mathcal{L}_{\text{latent}} = \frac{1}{N_{\text{L}}} \sum_{i,j\in E, i\neq j} \mathcal{L}_{\text{MSE}}\left(F^{\text{L}}_i, F^{\text{L}}_j\right), \tag{10}$$

where N_L is the number of $\mathcal{L}_{MSE}$ terms and E is the set of existing modalities. For example, if the MS modality is missing, E is {HR, SAR}. Next, the latent features of each modality pass through a router that masks out the features of missing modalities. In the router, the outputs of available modalities remain unchanged from their inputs. We obtain the latent features for the missing modalities by fusing the latent features of the available modalities. We employ simple averaging as the fusion method. Subsequently, the codebook of each modality, C_{HR}, C_{MS} and C_{SAR}, encodes the latent features by first applying a Gumbel softmax[93] to the latent features. We then perform matrix multiplication with the corresponding codebook. The modality codebook stores modality-specific features. Through this encoding, these features are embedded into the modality-specific latent features, which are then fed into decoders, D_{HR}, D_{MS} and D_{SAR}, to reconstruct the features of each modality, namely F^R_{HR}, F^R_{MS} and F^R_{SAR}. We calculate the feature-reconstruction loss $\mathcal{L}_{rec}$ between input features and the reconstructed features of available modalities as follows:

$$\mathcal{L}_{rec} = \frac{1}{N_R}\sum_{i\in E}\mathcal{L}_{MSE}\left(F_i, F_i^R\right), \tag{11}$$

where N_R is the number of $\mathcal{L}_{MSE}$ terms. The reconstructed features of each missing modality are used in the subsequent fusion. The reconstructed features of the available modalities are used only for the loss calculation and are not involved in the subsequent fusion. This process is formulated as follows:

$$F_i^M = \begin{cases} F_i, & \text{if } i \in E, \\ F_i^R, & \text{if } i \notin E. \end{cases} \tag{12}$$

Through the above process, SkySense++ achieves robust learning from labelled data in the RS-Sem dataset. The method is flexible enough to handle both training and testing samples if some modality data are unavailable.

Masked semantic recovery. To develop a principled learning objective that enables the network to predict categories of masked regions from the nearby unmasked regions, we designed the masked semantic recovery mechanism, which takes the fused multi-modal features F_{fus} as input and produces the bank prediction $P_B \in \mathbb{R}^{H\times W\times L}$, where H, W and L denote the height, width and number of channels in the predictions, respectively. The order of categories in bank prediction P_B corresponds to the order of categories in bank labels Y_B. As shown in Fig. 4c, random assignment changes the category index of 'river' from 1 to 3. Consequently, in the bank prediction, 'river' is assigned an index of 3. Unlike traditional multi-category segmentation methods, which generate predictions with a fixed category order, the masked semantic recovery module generates predictions with randomly shuffled category orders for each iteration, which aligns with the bank labels to handle unseen categories.

We adopted the segmentation head as the task head for masked semantic recovery when reconstructing the bank labels $Y_B \in \mathbb{R}^{H\times W\times L}$. In masked semantic integration, we encode category labels into masked semantic embeddings, which are randomly masked spatially with a randomly shuffled category order to serve as the network input. This approach forces the reconstruction of the bank labels in the mask semantic recovery to rely on nearby contextual regions rather than the order of the category labels, thereby enabling the in-context learning ability of the network. The semantic recovery loss $\mathcal{L}_{sem}$ is defined as

$$\mathcal{L}_{sem} = \mathcal{L}_{CE}(P_B, Y_B), \tag{13}$$

where $\mathcal{L}_{CE}$ represents the cross-entropy loss function. The total loss function during the semantic-enhanced pretraining is as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{sem} + \alpha\mathcal{L}_{latent} + \beta\mathcal{L}_{rec}, \tag{14}$$

where α and β are trade-off weights. In this study, we set both to 1.

Applications for EO tasks

SkySense++ supports a diverse range of downstream EO tasks. It can be applied to EO tasks in two distinct modes: the few-shot approach and the fine-tuning approach. The few-shot approach leverages a limited number of labelled samples as prompts to generate predictions without fine-tuning or retraining SkySense++. This capability makes it particularly suitable for time-sensitive EO tasks, for example, rapid flood monitoring. By contrast, the fine-tuning approach leverages further task-specific samples to fine-tune the entire network. This is suitable for applications that prefer high precision over a rapid response.

Few-shot approach. The few-shot capability of SkySense++ stems from its semantic-enhanced pretraining, where the in-context training is conducted through MSL. During MSL, the category labels are randomly masked and encoded through masked semantic integration. The training objective of the network is to recover the categories of masked regions from nearby contextual regions. Thus, SkySense++ can readily achieve few-shot capability when some prompt samples are available during testing. In typical few-shot scenarios, a few prompt images and their labels are considered references and used to generate predictions for a test image. Like semantic-enhanced pretraining, SkySense++ concatenates the prompt and test images as in-context image samples during few-shot inference. As there are no category labels for test images, we treat the category labels for the entire test image as being masked, which is then encoded by mask embedding. Subsequently, SkySense++ conducts mask semantic integration and mask semantic recovery to generate predictions for the test images from nearby contextual regions. The initial predictions generated by SkySense++ consist of L channels that are consistent with the length of the semantic bank. Another step is required to map the category order to the order of prompt labels based on the assignment of semantic banks in mask semantic integration.

Our few-shot approach supports two types of tasks: pixel-level semantic segmentation and image-level classification. As we employ pixel-level annotations for semantic-enhanced pretraining, SkySense++ naively supports few-shot segmentation tasks. For image-level classification tasks, we use image-level prompts, as shown in Fig. 3d, where prompt images for different categories are assigned to distinct prompt labels. SkySense++ generates pixel-level predictions through in-context learning. It then derives image-level predictions by computing the spatial proportion of each category. More details and experimental results are presented in Section Results.

Fine-tuning approach. SkySense++ also supports the fine-tuning approach for EO tasks. In this setting, SkySense++ uses sufficient downstream data to fine-tune the network parameters, allowing it to customize to the specific downstream EO task. The fine-tuning process boosts the performance of SkySense++ for a specific application scenario at the cost of using substantial labelled data and considerable offline computation. Therefore, this mode is suitable for non-urgent tasks with high accuracy requirements. For specific EO tasks, some modality data are not available. Therefore, we use only the encoders of the available modality in the task-specific data. This flexible and modular design ensures that SkySense++ can accommodate a wide range of EO tasks. During fine-tuning, an image x_{ft} is fed into the network, and after processing through SkySense++ plus a task-specific head, the prediction p_{ft} is obtained. The network parameters are optimized by minimizing the task-specific loss between the predictions and labels:

$$\mathcal{L}_{ft} = \mathcal{L}_{ft}(p_{ft}, y_{ft}), \tag{15}$$

where $\mathcal{L}_{ft}$ represents the fine-tuning loss and y_{ft} denotes the ground truth labels. Upon completion of fine-tuning, the model generates final predictions by performing inference on test images. More details and experimental results are presented in Section Results.

Data availability

The pretraining data and EO benchmarks used in this work are available via Zenodo at https://doi.org/10.5281/zenodo.14994429 (ref. 94). Following the collaborative agreement between Wuhan University and Ant Group regarding requirements for data redistribution compliance, we have checked the user agreements of the original data providers. Some prohibit redistribution of the data, for example, DeepGlobe, and some lack explicit redistribution guidelines, for example, Potsdam. Therefore, download links are available via Zenodo at https://doi.org/10.5281/zenodo.14994429 (ref. 94) for accessing these data. Interested researchers are required to sign the user agreements directly with the original data providers before accessing the datasets. Source data are provided with this paper.

Code availability

The code implemented in this work is available via GitHub at https://github.com/kang-wu/SkySensePlusPlus (ref. 95).

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Author contributions

K.W., Yingying Zhang, L.R., J.C., Yongjun Zhang and Y.L.: conceptualization. K.W., Yingying Zhang and L.R.: methodology, investigation, development, analysis, writing—original draft, writing—review and editing. Y.L., Yongjun Zhang and J.C.: supervision, methodology, analysis, writing—review and editing. J.W.: supervision, methodology and analysis. M.Y.: supervision, analysis, writing—review and editing. B.D., J. Lao, J. Luo and Q.Z.: analysis. K.W., L.Y., Z.Z., Y.S. and J.Z.: data curation. K.W., J. Luo and Q.Z. contributed to this work during their internships at Ant Group.

Competing interests

The authors declare no competing interests.

Additional information

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Extended Data Table 1 | Summary of our Earth observation benchmark, including the domains, task types, task datasets, modalities, ground sample distances (GSDs) and image sizes

Domain	Task type	Dataset	Modalities	GSD	Image size
Agriculture	Crop classification	Germany [33]	Sentinel-2*	10	24×24
Foresetry	Tree species classification	TreeSatAI-TS [45]	Airborne Sentinel-2* Sentinel-1*	0.2 10 10	304×304 6×6 6× 6
	Deforestation segmentation	Atlantic [35]	Sentinel-2	10	512×512
Oceanography	Oil spill segmentation	SOS [36]	Sentinel-1	10	256×256
Atmosphere	Air pollution regression	3pollution [37]	Sentinel-2 Sentinel-5P	10 2600	200×200 120×120
Biology	Wildlife detection	Kenya [38]	Airborne	-	3068×4603
Land surveying	LULC mapping	C2Seg-BW [39]	Gaofen-6 Gaofen-3	10 10	256×256 256×256
	Change detection	dsifn-cd [40]	GoogleEarth	0.3	512×512
Disaster management	Flood monitoring	Flood-3i [52, 53]	Airborne	0.05	256×256
		C2SMSFloods [41]	Sentinel-2 Sentinel-1	10 10	512×512 512×512
	Wildfire monitoring	CABUAR [42]	Sentinel-2	10	5490×5490
	Landslide mapping	GVLM [43]	GoogleEarth	0.3	1748×1748 ∼ 10808×7424
	Building damage assessment	xBD [44]	WorldView	0.3	1024×1024 1024×1024 1024×1024

Notes: '*' refers to time-series data.

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Extended Data Table 2 | Summary of collected datasets in our RS-Sem datasets, including dataset names, modalities, ground sample distances (GSDs), image sizes, number of categories, number of images and number of annotated pixels

Dataset	Modalities	GSD(m)	Image size	No. of categories	No. of images	No. of annotated pixels (B)
Five Billion Pixels [73]	Gaofen-2	4	6800×7200	24	150	7.34
Potsdam [74]	Airborne	0.05	6000×6000	5	38	1.36
Vaihingen [75]	Airborne	0.05	2494×2064	5	33	0.16
Deepglobe [76]	WorldView	0.5	2448×2448	6	1,146	6.86
iSAID [77]	Multiple Sensors	N/A	800×800 ∼ 4000×13000	15	2,806	27.27
LoveDA [78]	Spaceborne	0.3	1024×1024	7	5,987	6.27
DynamicEarthNet [79]	WorldView	0.3	1024×1024	7	54,750	57.40
	Sentinel-2*	10	32×32			
	Sentinel-1*	10	32×33			
Pastis-MM [28]	WorldView	0.3	1024×1024	18	2,433	40.81
	Sentinel-2*	10	32×32			
	Sentinel-1*	10	32×33			
C2Seg-AB [39]	Sentinel-2*	10	128×128	13	413	0.006
	Sentinel-1*	10	128×128			
FLAIR [81]	Spot-5	0.2	512×512	12	77,762	20.38
	Sentinel-2*	10	40×40			
DFC20 [82]	Sentinel-2	10	256×256	9	180,662	11.83
	Sentinel-1	10	256×256			
S2-naip [83]	NAIP	1	512×512	32	676,964	177.46
	Sentinel-2*	10	64×64			
	Sentinel-1*	10	64×64			
JL-16	Jilin-1	0.72	512×512	16	155,656	40.92
	Sentinel-1*	10	40×40			
Total	-	-	-	-	-	398.06

Notes: The number of categories excludes the background class. '*' refers to time-series data. 'N/A' indicates that the information is not publicly available.